

Module C — Front-End Development

Competition time

3 hours

Introduction

Background

Macau youth have access to a wide range of international skills competitions: WorldSkills, Microsoft Office Specialist (MOS) World Championship, Adobe Certified Professional (ACP) International Certification, Huawei ICT Competition, Innovation and Technology Elite Competition, and more — spanning IT, creative design, engineering, and hospitality. These competitions offer young people in Macau the chance to showcase their talents, earn internationally recognised certifications, and engage with peers from around the world.

Government departments and institutions also organise free training workshops, hands-on experience sessions, and selection briefings to develop vocational skills in young people.

However, these opportunities are scattered across different channels — individual organisation websites, social media pages, press releases. Young people often struggle to know what competitions are open for registration, which ones match their interests, or what free training is available. This fragmentation means many talented young people miss out.

The Platform

Skills Macau is an information hub designed to solve this problem. It brings together competitions, training events, and skill development resources from Macau's government bodies and institutions into one place, allowing young people to discover:

- What international competitions are available?
- What free training, workshops and selection briefings can they join?
- What skill development opportunities does each organisation offer?

Your Task

In this module, you will build the front-end application for the Skills Macau platform using a front-end framework. All content data is provided in `data.json` (organisations, skills, competitions, events, and Macau competition milestones). No backend or database is required. Assessment focuses on component architecture, state management, routing, responsive layout, accessibility, and client-side interactivity.

Provided Resources

Resource	Description
<code>data.json</code>	All page data (organizations, skills, competitions, events, milestones)
<code>assets/</code>	Images (organization logos, skill icons, competition covers, milestone photos)

Technical Requirements

You **must** use a front-end framework:

Framework	Notes
React	With Vite; React Router
Vue 3	With Vite; Vue Router

CSS: Use **Tailwind CSS** or **Bootstrap** (pre-installed in the competition environment).

Global Requirements

The website must support three breakpoints: Mobile (< 768px), Tablet (768px–1024px), and Desktop (> 1024px). All interactive elements must be keyboard-operable. Images must have **alt** attributes. Dynamic content must use appropriate ARIA attributes. Color contrast must meet WCAG AA.

All page content is loaded from **data.json**. The application must be a Single Page Application using client-side routing — no full page reloads. Use Vite as the build tool; **npm run build** must produce static files successfully.

Pages & Features

1. Navigation

The navigation bar appears on every page. On desktop, it displays horizontally: the Skills Macau logo, followed by links to Home, Competitions, Events, and Timeline. The link for the current page must be visually highlighted. The nav bar stays fixed at the top of the viewport when scrolling.

On mobile, the nav collapses into a hamburger menu that toggles open and closed.

2. Home Page

The home page opens with a full-width **Hero section** — a large visual area with the platform name "Skills Macau", a tagline, and two call-to-action buttons linking to Competitions and Events. The hero's background, typography, color scheme, and button styling are yours to design.

Below the hero, a **stats bar** shows three numbers computed from **data.json**: the total number of organizations, the total number of competitions, and the number of upcoming events.

A **Latest Competitions** section displays the 3 most recently created competitions as cards. Each card shows the cover image, name, organization, date range, and a status tag.

An **Upcoming Events** section displays the next 3 events (date ≥ today, sorted by date ascending). Each card shows the name, date, type tag, and location.

A **footer** appears at the bottom of every page with copyright information and a back-to-top button.

3. Competitions

List Page. Competitions are displayed in a card grid — 3 columns on desktop, 2 on tablet, 1 on mobile. Each card shows the cover image, name, organization, date range, status tag, and up to 3 related skill tags (with a "+N" overflow indicator for the rest).

Users can filter competitions in three ways, all client-side without page reload:

- **Status** — a button group toggling between All, Upcoming, Ongoing, and Completed
- **Organization** — a dropdown whose options are dynamically generated from the data
- **Skill** — a multi-select tag cloud; selecting multiple skills applies intersection filtering

A result count is shown (e.g. "Showing 5 / 20 competitions"). Competitions can be sorted by date (newest/oldest) or by name (A–Z / Z–A). When no competitions match the filters, a friendly empty state message is displayed.

Detail Page. Clicking a competition card navigates to `/competitions/:competitionId`. The detail page shows a breadcrumb (Home > Competitions > name), a full-width cover image, the competition name, organization (as plain text), date range, status tag, and full description. Related skill tags are clickable and navigate to the corresponding skill detail page. A back button returns to the list, and previous/next navigation allows browsing between competitions.

4. Skill Detail

Each skill has a dedicated page at `/skills/:skillId`. It shows a breadcrumb (Home > Skills > name), the skill icon, name, category, and full description.

Below the description, two sections are displayed:

- **Competitions with this skill** — a list of cards (name, date, status), each linking to the competition detail page
- **Related events** — a list of items (name, date, type), each linking to the event detail page

When a skill has no related competitions or events, an empty state message is shown. Navigating to an invalid skill ID displays a friendly message.

5. Events

List Page. Events are displayed in a card grid with the same responsive rules as competitions. Each card shows the name, organization, skill name, type tag, date, time, and location.

A search bar at the top filters events by name as the user types, with a 300ms debounce. A type dropdown filters by event type (Workshop, Lecture, Training, Experience, Selection Briefing). Past events (date < today) are visually distinguished with reduced opacity and shown by default; a toggle allows hiding them. Events are sorted by date ascending.

Detail Page. Clicking an event card navigates to the detail page. It shows a breadcrumb, the event name, organization (plain text), skill (clickable, linking to the skill detail page), type tag, date and time range, and location. The fee field displays "Free" when empty. The full description is shown, and a back button returns to the list.

6. ★ Featured Competition — WorldSkills Timeline

To give users a deeper look at one of the platform's major competitions, a dedicated Timeline page traces the history of WorldSkills and Macau's participation from 1985 to the present. Milestones are stored in the `milestones` array in `data.json`; each milestone has a `competitionId` field. Filter milestones by `competitionId: "ws48"` to display the 8 WorldSkills milestones in chronological order.

Milestones are displayed in chronological order. Each shows the year, title, description, and image (if available in the data). Images must use lazy loading.

The `ws48` competition object in `data.json` includes a `gallery` array of photo paths from past WorldSkills events. Display these photos in a **photo gallery** section on the Timeline page. Clicking a thumbnail opens the full image in a lightbox or modal overlay.

The visual presentation is yours to design — vertical timeline, horizontal timeline, card cascade, interactive map, or any other format you choose. The page must include:

- At least one **scroll-triggered animation** — elements animate into view as the user scrolls
- At least one **interactive feature** — such as expand/collapse, filter, or zoom (the photo gallery lightbox qualifies)

Architecture Requirements

Components

Reusable UI elements — cards, buttons, tags, filter bars, breadcrumbs, the footer — must be extracted as independent components. Pages are composed from these components, not written as single monolithic blocks.

Routing

Use a front-end router. Every page must have a corresponding route. Skill and competition detail pages use dynamic route parameters (`/skills/:skillId`, `/competitions/:competitionId`). Unknown routes display a friendly 404 page.

State Management

Data loaded from `data.json` must be stored in a global state accessible by all components. Filter, sort, and search state must be correctly maintained. Avoid prop drilling.

Error Handling

If `data.json` fails to load, show an error message with a retry button. If an image fails to load, show a placeholder. If a dynamic route parameter does not match any data, show a friendly message rather than a blank page or crash.

Work Organization

You must initialise a **Git repository** in your project folder and make at least one commit during development. A **.gitignore** file must be present (Vite templates include one by default — ensure **node_modules/** and **dist/** are ignored).

Include a **README.md** that documents the project name and the commands to run the dev server and production build.

Submission

- Package the complete project as **XX_Module_C.zip** (XX = seat number)
 - Must include: **package.json**, all source code, **data.json**, **assets/** directory
 - Do **not** include **node_modules/** (judges will run **npm install && npm run build**)
 - Judges will test the built static files in Google Chrome desktop
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Assessment

Module C will be assessed using Google Chrome. Marking follows the WorldSkills Occupation Standards (WSOS) with Measurement (pass/fail) and Judgement (0–3 scale) criteria.

Mark Distribution

Section	Marks
Work Organization (build, git, README)	6
Navigation	6
Home Page	15
Competitions List & Detail	17
Skill Detail Page	8
Events List & Detail	12
Timeline Page	16
Architecture & Code Quality	10
Accessibility & Overall Design	10
Total	100